

Pellevoisin: the Raspberry Pi 4 header

Pin 1 top left with the board's edge at the top: odd pins in the inner row, even pins in the outer row next to the edge. Highlighted cells are this organ's connections. Every GND pin is the Pi's ground, and nothing on the organ's side ever touches this header.

ours	function	pin	pin	function	ours
MIDI: 6N137 module IN+	3.3 V	1	2	5 V	free (Pi fed by USB-C from the SD-25A-5)
free; I2C stays off	GPIO 2 / SDA	3	4	5 V	free
ON/OFF SWITCH, one side	GPIO 3 / SCL	5	6	GND	ON/OFF SWITCH, other side
	GPIO 4	7	8	GPIO 14 / TXD	MIDI: 6N137 module IN-
RESET opto cathode (PC817 pin 2)	GND	9	10	GPIO 15 / RXD	free
RESET: 330 ohm to PC817 anode	GPIO 17	11	12	GPIO 18	
STATUS LED: 330 ohm to LED anode	GPIO 27	13	14	GND	STATUS LED cathode
	GPIO 22	15	16	GPIO 23	
	3.3 V	17	18	GPIO 24	PUMP: SSR terminal 3 (+)
	GPIO 10	19	20	GND	PUMP: SSR terminal 4 (-)
	GPIO 9	21	22	GPIO 25	
	GPIO 11	23	24	GPIO 8	
	GND	25	26	GPIO 7	
never	ID_SD	27	28	ID_SC	never
	GPIO 5	29	30	GND	
	GPIO 6	31	32	GPIO 12	
	GPIO 13	33	34	GND	
	GPIO 19	35	36	GPIO 16	
	GPIO 26	37	38	GPIO 20	
	GND	39	40	GPIO 21	

Each line is paired with its nearest ground so every cable is a short twisted pair: 5 and 6 for the switch, 11 and 9 for reset, 13 and 14 for the status LED, 18 and 20 for the pump. The MIDI pair is pins 1 and 8 and carries no ground: TX is its return. Pins 3 and 5 are the I2C bus, which stays disabled so GPIO 3 is free for the switch and wakes a halted Pi when pulled low. The DSI screen uses its own ribbon connector.

Part pinouts: PC817 pin 1 anode, 2 cathode, 3 emitter, 4 collector (dot at pin 1). SSR-xx DA: 1 and 2 load (mains), 3 positive and 4 negative input, 3 to 32 V DC. 6N137 module: IN+ and IN- marked on the terminals; VCC, OUT, GND on the other side. SD-25A-5: +Vin/-Vin from the 12 V bus through a 2 A fuse, +V/-V trimmed to 5.1 V at the Pi into its USB-C; 1500 V isolation inside. organ_web: --pump 24 --power-switch 3; reset on 17. Status LED: config.txt dtoverlay=gpio-led,gpio=27,label=pellevoisin,trigger=heartbeat; it beats while Linux runs and is dark when halted.

Pellevoisin: the box, wired

Two halves in one aluminium case, and exactly four things cross between them, three by light and one by a transformer: the MIDI opto, the reset opto, the SSR, and the isolated converter that makes the Pi's 5 V from the 12 V bus.

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MAINS -- IEC inlet with fuse +- E ---- earth stud on the case ---- motor frame
+- N -----+----- motor N
|
|          PSU N
+- L -- PANEL SWITCH, 2 poles, 3 positions, centre off, motor rated
      pole A common +- BOOK -----> motor L
                    +- AUTO -> SSR 1 -- SSR 2 -----> motor L
      pole B common --- AUTO -> PSU L   (its on pin strapped to +12 V: on with Auto
      (OFF: neither pole connected; the supply's fan cools the box in Auto)

ORGAN SIDE (the 12 V supply's ground)
PSU 12 V, several wires --> 12 V bus bar
PSU GND, several wires ---> organ GND bus bar
PSU on pin -----> PSU +12 V: on with Auto
12 V bus / GND bus -----> boards 1-4, 5 A fuses
12 V bus -[2 A fuse]-> SD-25A-5 +Vin, GND bus > -Vin
----- transformer inside the SD-25A-5 -----
last board ISP pin 2 -----> RJ12 pin 6 = 5 V bus
RJ12 pin 1 = /RESET bus (jumped board to board)
RJ12: MIDI signal, GND, as shipped
6N137 module: VCC < RJ12 pin 6, GND < RJ12 GND,
              OUT > boards' MIDI in
PC817: collector - RJ12 pin 1, emitter - RJ12 GND
SSR 1 / 2: on the mains side, above; MOV across them

PI SIDE (the converter's output ground)
SD-25A-5 +V / -V, trimmed to 5.1 V --USB-C--> Pi 4
DSI ribbon --> touchscreen
GPIO 3 (pin 5) -- antique switch -- Pi GND (pin 6)   on/off
GPIO 14 TX (pin 8) -- 6N137 IN-                      MIDI
3.3 V (pin 1) ----- 6N137 IN+
GPIO 24 (pin 18) -- SSR 3 (+)                         pump
Pi GND (pin 20) --- SSR 4 (-)
GPIO 27 (pin 13) -[330 ohm]- LED - Pi GND (pin 14) alive
GPIO 17 (pin 11) -[330 ohm]- PC817 anode               reset
Pi GND (pin 9) ----- PC817 cathode
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Rules

- No wire, bar or chassis lug between the organ's GND bus and the Pi's ground: the Pi's 5 V comes through the converter's transformer, never from a buck. The earth stud carries earth only, and no DC negative goes to it.
- Mains bus bars shrouded, or covered DIN terminal blocks instead. Low-voltage bars may be bare.
- Mains and low voltage on opposite sides of a partition: inlet, switch, supplies' inputs, SSR 1 and 2, motor lead on one side; Pi, RJ12 bus, optos, SSR 3 and 4 on the other. A wire that must cross does so once, at right angles.
- The SSR's heat sink in the 12 V supply's fan stream, thermal paste under the module. A varistor across SSR 1 and 2.
- Only the live is switched, ever. Neutral and earth run straight through. The panel switch is motor rated: 1 HP at 125 V, or 20 A. The antique on/off switch carries microamps and may be anything.
- Book: the motor runs, nothing else has power. Auto: the 12 V supply (Lenovo SP50H29523, 15 A) starts at once, its on pin strapped to +12 V, and the bus, the fan, the boards and the Pi through its converter come up together; the motor runs only when the SSR says so. Off: everything isolated (the SSR alone leaks a milliamp).
- Before the panel goes from Auto to Off: the antique switch off, or Shut down on the Service tab, and wait for the status LED to go dark.